

Aggregate Functions, Multiset Relational Algebra, and Database Modification

1 Aggregate Functions and Operations

1.1 Definition

An **aggregate function** takes a collection of values and returns a single value as a result.

- **avg**: average value
- **min**: minimum value
- **max**: maximum value
- **sum**: sum of values
- **count**: number of values

1.2 General Aggregate Operation in Relational Algebra

Let:

- E : any relational-algebra expression
- $G_1, G_2, \dots, G_n$: list of attributes on which to group (can be empty)
- F_i : aggregate function
- A_i : attribute name

Notation:

$$G_1, G_2, \dots, G_n \ F_1(A_1), F_2(A_2), \dots, F_k(A_k) \ (E)$$

Some books use γ instead of calligraphic G .

Example

Relation r :

A	B	C
α	α	7
α	β	7
β	α	3
β	β	10
α	β	0
β	β	0
β	α	0
β	β	0

Aggregate:

$$\text{sum}(C)(r) = 27$$

1.3 Example: Average Salary per Department

`dept_name avg(salary)(instructor)`

SQL:

```
SELECT dept_name, AVG(salary) AS avg_salary
FROM instructor
GROUP BY dept_name;
```

1.4 Distinct in Aggregates

Some queries require eliminating duplicates before aggregation.

Example:

`count-distinct(ID)( $\sigma_{\text{semester}='Spring' \wedge \text{year}=2010}(\text{teaches})$ )`

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2 Multiset Relational Algebra

2.1 Definition

- **Set**: only one copy of each tuple.
- **Multiset (bag)**: multiple occurrences of the same tuple allowed.

2.2 Differences from Pure Relational Algebra

- **Selection**: keeps all duplicates that satisfy condition.
- **Projection**: keeps duplicates; one tuple per input tuple.
- **Cross Product**: if t_1 occurs m times and t_2 occurs n times, $t_1.t_2$ occurs $m \times n$ times.
- **Union**: $m + n$ copies.

- **Intersection:** $\min(m, n)$ copies.
- **Difference:** $\max(0, m - n)$ copies.

2.3 SQL Equivalence

Without aggregation:

$$\pi_{A_1, \dots, A_n}(\sigma_P(r_1 \times \dots \times r_m))$$

With aggregation:

$$A_1, A_2 \text{ sum}(A_3)(\sigma_P(r_1 \times \dots \times r_m))$$

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3 Modification of the Database

3.1 Deletion

Relational algebra:

$$r \leftarrow r - E$$

Example

Delete all account records in the Perryridge branch:

$$\text{account} \leftarrow \text{account} - \sigma_{\text{branch_name}='Perryridge'}(\text{account})$$

3.2 Insertion

Relational algebra:

$$r \leftarrow r \cup E$$

Example

Insert tuple ("A-973", "Perryridge", 1200) into account:

$$\text{account} \leftarrow \text{account} \cup \{("A-973", "Perryridge", 1200)\}$$

3.3 Updating

Uses **generalized projection** to change attribute values.

Example

Increase all balances by 5%:

$$\text{account} \leftarrow \pi_{\text{account_number}, \text{branch_name}, \text{balance} \times 1.05}(\text{account})$$

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4 Example Queries

- Customers who have both a loan and an account:

$$\pi_{\text{customer_name}}(\text{borrower}) \cap \pi_{\text{customer_name}}(\text{depositor})$$

- Customers with accounts at all branches in Brooklyn:

$$\pi_{\text{customer_name}, \text{branch_name}}(\text{depositor} \bowtie \text{account}) \div \pi_{\text{branch_name}}(\sigma_{\text{branch_city}='Brooklyn'}(\text{branch}))$$